

Pavan Venkata Manjunath Mallipudi

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EDUCATION

Arizona State University

B.S., Computer Science | Data Science Minor (GPA: 3.76)

Aug 2022 - May 2026

Tempe, AZ

- **Achievements:** Magna Cum Laude, 5x Dean's List
- **Coursework:** Machine Learning, Data Visualization, Probability and Statistics, Statistical Modeling and Inference, Data in R & Python
- **Certifications:** AWS AI Practitioner, AWS Cloud Practitioner

TECHNICAL SKILLS

Programming Languages: Python, SQL, R

Data Engineering & Platforms: PostgreSQL, MySQL, Apache Spark (PySpark), ETL/ELT Pipelines, Data Modeling, Dimensional Modeling, Data Warehousing, Databricks, AWS, Docker, Git/GitHub, dbt, Snowflake, Airflow

Data & BI Technologies: Pandas, NumPy, Excel, Power BI, Tableau, Tableau Prep, Streamlit, Plotly, Matplotlib, Jupyter Notebook

Data Analytics: Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization, A/B Testing, Cohort Analysis, KPI Reporting, Ad Hoc Analysis, Root Cause Analysis, Stakeholder Management, Database Management, Hypothesis Testing

EXPERIENCE

ASU Social Embeddedness | Community Engagement Data Analyst

Sep 2024 - May 2026

- Administered and validated **2,300+** community-engagement activities across **32 ASU units** and **650 community organizations**, implementing standardization and validation rules that **increased required-field completeness from 78% to 96%**.
- Built Tableau Prep pipelines processing **30,000+** activity-unit-partner rows and developed **6 interactive dashboards** tracking **10 institutional KPIs**, used across 32 ASU units, **reducing monthly reporting prep from 4 days to 6 hours**.
- Delivered **24 platform demonstrations** and support sessions to **70+ faculty and staff** across **18 departments**, contributing to a **28% increase** in active platform contributors and a **35% reduction in recurring data-entry errors**.

Food Forest AI | Data Analyst Intern

Jun 2025 - Jul 2025

- Built a document-ingestion pipeline converting **650+** supplier PDFs into standardized JSON profiles across **15 attributes**, including extrusion capabilities, hot/cold-fill processing, facility locations, and SQF/Organic certifications.
- Normalized 40+ variations in processing capabilities and certification naming through data-cleaning logic, **reducing average supplier onboarding time from 12 to 4 minutes**.
- Built Python/SQL data-quality checks across **2,500+ supplier profiles** (420 missing fields, 275 formatting anomalies, 110 duplicates found) and Power BI dashboards tracking **8,000+ B2B search events** across 5 pilot clients to support product decisions.

DigiClips (Capstone) | Data Engineer

Aug 2025 - Apr 2026

- Automated nightly retention cleanup using MySQL Event Scheduler, purging ~20,000 expired metadata records monthly and **cutting manual maintenance from 3 hours to 30 minutes per week**.
- Built data-quality routines across 12 MySQL tables, **removing 4,500 duplicate records** and correcting 2,000 timestamp inconsistencies, with automated weekly database dumps for backup and retention compliance.
- Refactored 3 search stored procedures after reverse-engineering the relational schema, **cutting median query latency 64% (1.8s to 650ms)** on a 500K-record test database.

PROJECTS

Subscription Churn Analytics | SQL, dbt, DuckDB, Python, ECharts | [Deployed Project](#)

- Built a dbt + DuckDB warehouse (16 models, 21 tests) over **23M KKBox billing transactions** and 12.1M renewal decisions, producing MRR bridges that reconcile every month, cohort retention, and NRR / GRR of 77.7% / 76.0%.
- Found billing behavior drives churn more than listening: **manual renewers churn at 20.9% vs 2.8% on auto-renew (7.5x)** and 50%+ discount periods at 69.2%, quantifying NT\$79.0M of monthly revenue churned over 12 months.
- Caught data-quality issues that changed the answers, including 23 missing payment-method months that faked a 14% churn spike and a survivor bias that **overstated 12-month retention by about 20 points**.

ReadmitScope US | Python, Pandas, SciPy, CMS Data, React | [Deployed Project](#)

- Found readmission performance is a hospital-level trait: **9.2% of hospitals** reporting 3+ conditions are worse than expected on every one, about twice the 4.7% chance predicts, across 11,720 CMS measures and 6 conditions.
- Built reproducible Python (Pandas) cleaning and QA workflows for suppression-prone CMS data, then used hospital-clustered tests (SciPy) to show the "small hospitals do worst" pattern is **a reporting artifact (p=0.76)** while star rating tracks readmissions (p<0.001).
- Designed and deployed an interactive React dashboard with a searchable, sortable 2,833-hospital explorer and state rankings, translating CMS quality metrics into facility-level comparisons and actionable insights.

Credit Decisioning Engine | Python, SQL (DuckDB), XGBoost, SHAP | [Deployed Project](#)

- Analyzed **618K LendingClub loans** using Python and DuckDB SQL to evaluate credit risk, repayment outcomes, and approval strategy, finding the right policy **depends on cost of capital**: nearly everyone should be approved at a 0% funding cost, but not at 4%.
- Built a profit-based approval strategy showing that, at a 4% cost of funds, declining the riskiest 12% increases profit 13.8%, and that ranking applicants with the model **earns \$6.4M more** than LendingClub's sub-grade at 84% approval.
- Developed fairness and drift monitoring using income-band analysis and PSI in an interactive dashboard, finding that at an 80% approval stress test applicants under \$40K are approved at **0.53x the top income band's rate**.